

Лабораторная 3

Тема: Настройка виртуальной локальной сети (VLAN)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN

-> две сети VLAN, значит логично будет каждой сети назначить свою IP-сеть:

Пример назначения адреса:

```
PC2> ip 192.168.2.20/24
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
. done
```

Итого:

```
PC1  192.168.1.10/24  VLAN 20
PC3  192.168.1.30/24  VLAN 20
PC5  192.168.1.50/24  VLAN 20

PC2  192.168.2.20/24  VLAN 333
PC4  192.168.2.40/24  VLAN 333
PC6  192.168.2.60/24  VLAN 333
```

-> Затем на всех коммутаторах создал обе VLAN:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
```

```

Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#end
Switch#
*Aug  4 09:25:25.505: %SYS-5-CONFIG_I: Configured from console by console
Switch#write memory
Building configuration...
Compressed configuration from 5213 bytes to 1976 bytes[OK]
*Aug  4 09:25:48.034: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...
Switch#
*Aug  4 09:25:49.313: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Switch#show vlan brief

```

VLAN	Name	Status	Ports
----	-----	-----	-----
1	default	active	Gi2/0
20	VLAN20	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

-> Удалил, VLAN100, VLAN0200, VLAN0300 чтобы не мешали

```

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no vlan 100
Switch(config)#no vlan 200
Switch(config)#no vlan 300
Switch(config)#end
Switch#write memory
*Aug  4 09:26:54.827: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
Compressed configuration from 5149 bytes to 1939 bytes[OK]
Switch#
*Aug  4 09:27:05.131: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...
*Aug  4 09:27:06.099: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.

```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi2/0
20	VLAN20	active	
333	VLAN333	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

-> Проверил STP, он включен, но не настроен:
результат вывода для Layer2Switch2

```
Switch#show spanning-tree summary
```

```
Switch is in pvst mode
```

```
Root bridge for: VLAN0001, VLAN0020, VLAN0333
```

```
Extended system ID is enabled
```

```
Portfast Default is disabled
```

```
PortFast BPDU Guard Default is disabled
```

```
Portfast BPDU Filter Default is disabled
```

```
Loopguard Default is disabled
```

```
EtherChannel misconfig guard is enabled
```

```
Configured Pathcost method used is short
```

```
UplinkFast is disabled
```

```
BackboneFast is disabled
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	9	9
VLAN0020	0	0	0	8	8
VLAN0333	0	0	0	8	8

```
3 vlans
```

-> на Layer2Switch2:

```
Switch#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Switch(config)#spanning-tree vlan 20 priority 8192
```

```
Switch(config)#spanning-tree vlan 333 priority 4096
```

```
Switch#write memory
```

```
Building configuration...
```

```
Compressed configuration from 5223 bytes to 1987 bytes[OK]
```

```
Switch#
```

```
*Aug 4 09:43:54.094: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
```

updated on disk. Please wait...

*Aug 4 09:43:55.100: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Switch#show spanning-tree bridge

Vlan	Bridge ID	Hello Time	Max Age	Fwd Dly	Protocol
VLAN0001	32769 (32768, 1) 0c1a.cff5.0000	2	20	15	ieee
VLAN0020	8212 (8192, 20) 0c1a.cff5.0000	2	20	15	ieee
VLAN0333	4429 (4096, 333) 0c1a.cff5.0000	2	20	15	ieee

-> на Layer2Switch1:

Switch#show spanning-tree bridge

Vlan	Bridge ID	Hello Time	Max Age	Fwd Dly	Protocol
VLAN0001	32769 (32768, 1) 0c50.955b.0000	2	20	15	ieee
VLAN0020	4116 (4096, 20) 0c50.955b.0000	2	20	15	ieee
VLAN0333	8525 (8192, 333) 0c50.955b.0000	2	20	15	ieee

-> таким образом:

- Layer2Switch-1 - root для VLAN 20
- Layer2Switch-2 - root для VLAN 333
- каждый из них будет резервным для другого VLAN
- Layer2Switch-3, 4 и 5 сохраняют стандартный приоритет 32768

-> Далее настройка IEEE 802.1Q, подробный вывод с SW1:

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#\$range GigabitEthernet0/0 - 3 , GigabitEthernet1/0 - 3

Switch(config-if-range)# switchport trunk encapsulation dot1q

Switch(config-if-range)# switchport mode trunk

Switch(config-if-range)# switchport trunk native vlan 333

Switch(config-if-range)# switchport trunk allowed vlan 20,333

Switch(config-if-range)#end

Switch#write memory

*Aug 5 02:37:19.529: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...

Compressed configuration from 6286 bytes to 2372 bytes[OK]

Switch#

```
*Aug  5 02:37:31.082: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
```

```
*Aug  5 02:37:32.099: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

```
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	333
Gi0/3	on	802.1q	trunking	333
Gi1/0	on	802.1q	trunking	333
Gi1/1	on	802.1q	trunking	333
Gi1/2	on	802.1q	trunking	333
Gi1/3	on	802.1q	trunking	333

```
Port          Vlans allowed on trunk
```

Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333
Gi0/3	20,333
Gi1/0	20,333
Gi1/1	20,333
Gi1/2	20,333
Gi1/3	20,333

```
Port          Vlans allowed and active in management domain
```

Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333
Gi0/3	20,333
Gi1/0	20,333
Gi1/1	20,333
Gi1/2	20,333
Gi1/3	20,333

```
Port          Vlans in spanning tree forwarding state and not pruned
```

Gi0/0	none
Gi0/1	none
Gi0/2	20,333
Gi0/3	none
Gi1/0	none
Gi1/1	none
Gi1/2	20,333
Gi1/3	20,333

```
Switch#
```

-> на SW3, SW4, SW5 портов меньше (на примере SW3):

```

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range GigabitEthernet0/0 - 3
Switch(config-if-range)# switchport trunk encapsulation dot1q
Switch(config-if-range)# switchport mode trunk
Switch(config-if-range)# switchport trunk native vlan 333
Switch(config-if-range)# switchport trunk allowed vlan 20,333
Switch(config-if-range)#end
Switch#write memory
Building configuration...
Compressed configuration from 5486 bytes to 2117 bytes
*Aug  5 02:41:38.758: %SYS-5-CONFIG_I: Configured from console by console
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
867
[OK]
*Aug  5 02:41:57.136: %SYS-3-CPUHOG: Task is running for (1997)msecs, more
than.
*Aug  5 02:41:57.524: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updat.
Switch#
Switch#
*Aug  5 02:41:58.860: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to.
Switch#
Switch#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	333
Gi0/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333
Gi0/3	20,333

Port	Vlans allowed and active in management domain
Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333
Gi0/3	20,333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333

Gi0/3 none

-> после настройки vlan и 802.1Q:

Switch>show spanning-tree vlan 20

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 4116
 Address 0c50.955b.0000
 Cost 4
 Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
 Address 0cb9.c0de.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 15 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----

Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr

-> пример вывода на SW5:

Switch#show interface trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	on	802.1q	trunking	333
Gi0/2	on	802.1q	trunking	333
Gi0/3	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20,333
Gi0/1	20,333
Gi0/2	20,333
Gi0/3	20,333

Port	Vlans allowed and active in management domain
Gi0/0	20,333
Gi0/1	20,333

```

Gi0/2      20,333
Gi0/3      20,333

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0      20
Gi0/1      none
Gi0/2      333
Gi0/3      none

```

-> Итого сейчас:

- Межкоммутаторные порты работают через 802.1Q.
 - Native VLAN установлена в 333.
 - Через trunk разрешены только VLAN 20,333.
 - STP выбрал разные корневые коммутаторы для двух VLAN.
 - Избыточные пути заблокированы.
- > Теперь надо подключить компьютеры

```
Switch#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	1	auto	auto	unknown
Gi1/1		connected	1	auto	auto	unknown

->Пример подключения PC1 к vlan 20 через access порт

```

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface GigabitEthernet1/0
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#end
Switch#
*Aug  5 09:19:26.311: %SYS-5-CONFIG_I: Configured from console by console
Switch#

```



```
Switch#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	1	auto	auto	unknown

-> аналогично с vlan 333

```
Switch#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#interface GigabitEthernet1/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 333
```

```
Switch(config-if)# spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops. Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only have effect when the interface is in a non-trunking mode.

```
Switch(config-if)#end
```

```
Switch#write memory
```

Building configuration...

Compressed configuration from 5595 bytes to 2154 bytes

*Aug 5 09:34:24.994: %SYS-5-CONFIG_I: Configured from console by console[OK]

```
Switch#
```

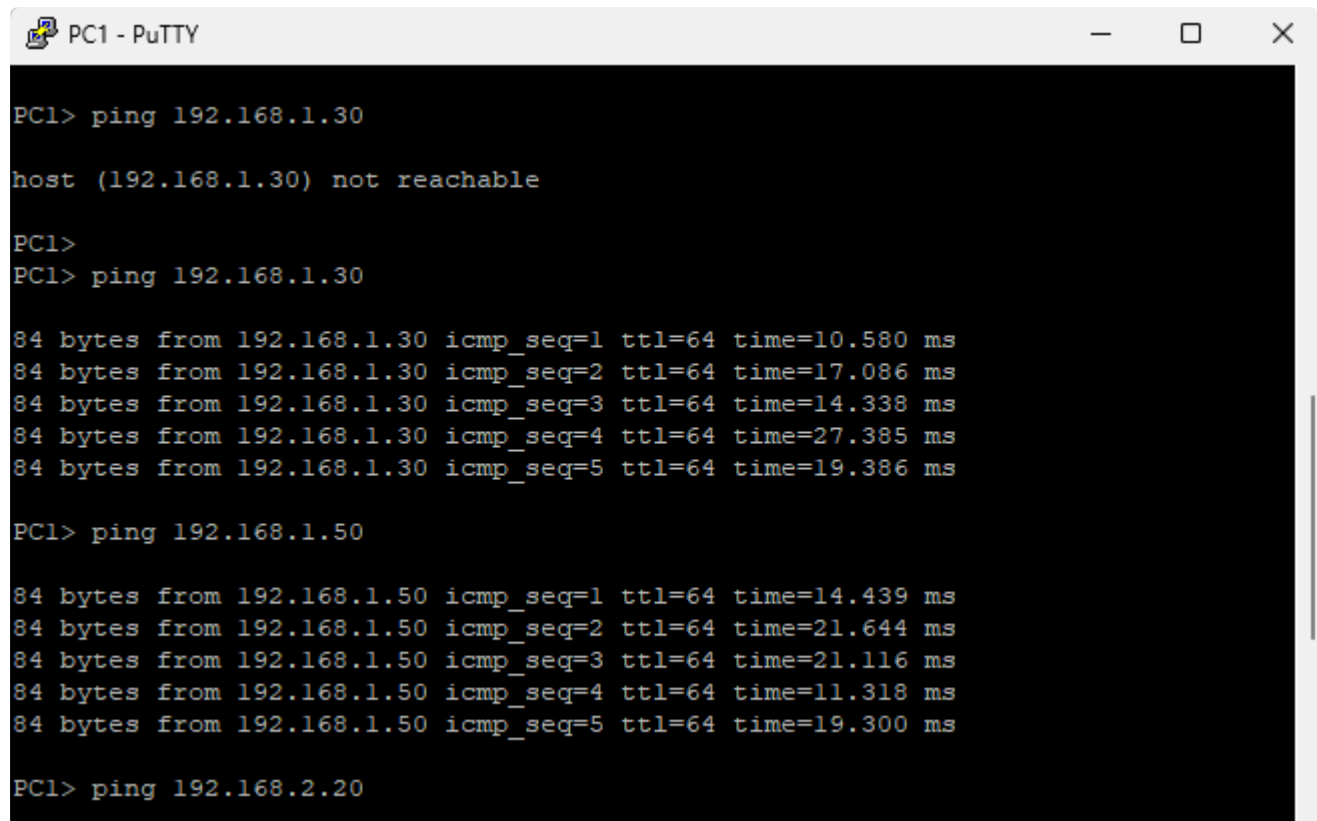
*Aug 5 09:34:35.172: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Aug 5 09:34:36.328: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

```
Switch#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать



```
PC1 - PuTTY
PC1> ping 192.168.1.30
host (192.168.1.30) not reachable
PC1>
PC1> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=10.580 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=17.086 ms
84 bytes from 192.168.1.30 icmp_seq=3 ttl=64 time=14.338 ms
84 bytes from 192.168.1.30 icmp_seq=4 ttl=64 time=27.385 ms
84 bytes from 192.168.1.30 icmp_seq=5 ttl=64 time=19.386 ms
PC1> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=14.439 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=21.644 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=21.116 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=11.318 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=19.300 ms
PC1> ping 192.168.2.20
```

```
PC1> ping 192.168.1.30
```

```
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=10.580 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=17.086 ms
84 bytes from 192.168.1.30 icmp_seq=3 ttl=64 time=14.338 ms
84 bytes from 192.168.1.30 icmp_seq=4 ttl=64 time=27.385 ms
84 bytes from 192.168.1.30 icmp_seq=5 ttl=64 time=19.386 ms
```

```
PC1> ping 192.168.1.50
```

```
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=14.439 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=64 time=21.644 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=64 time=21.116 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=64 time=11.318 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=64 time=19.300 ms
```

```
PC1> ping 192.168.2.20
```

```
No gateway found
```

```
PC1> ping 192.168.2.40
```

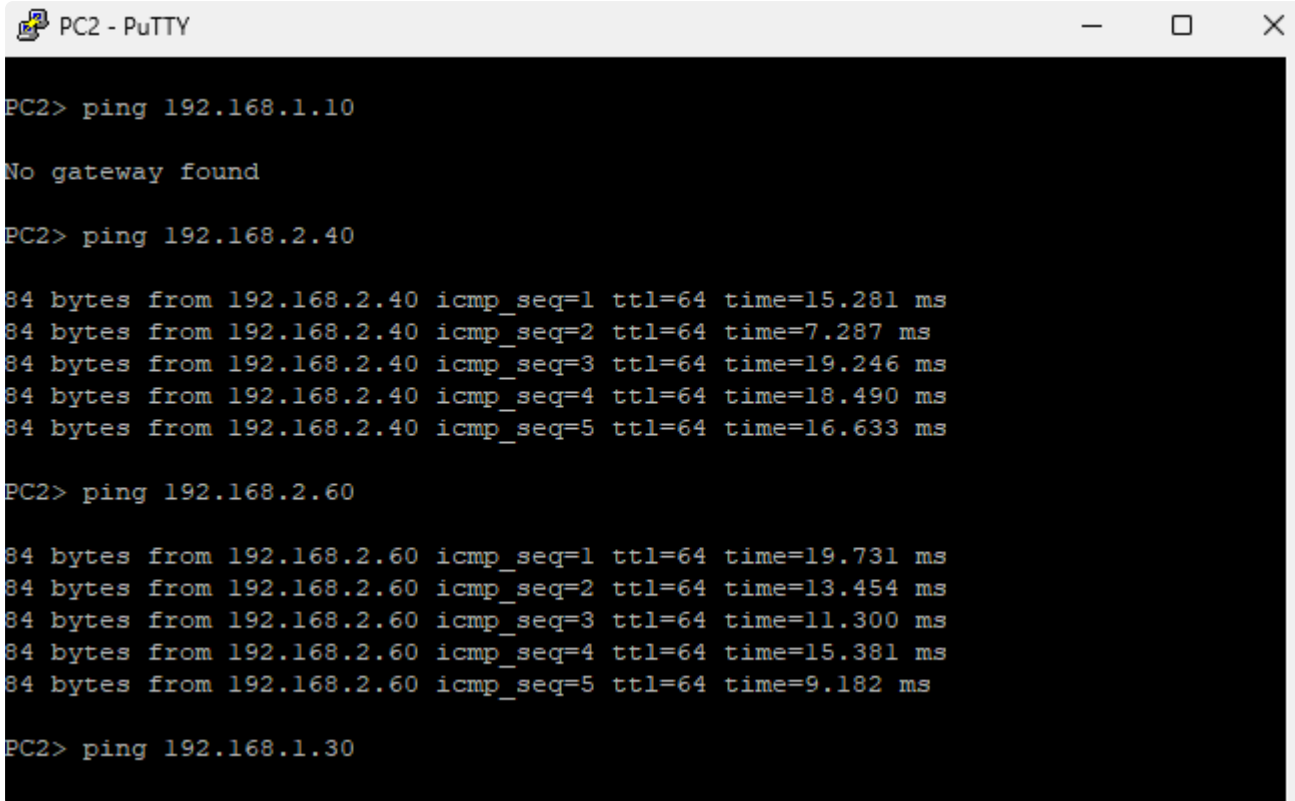
```
No gateway found
```

```
PC1> ping 192.168.2.6
```

No gateway found

```
PC1> ping 192.168.2.60
```

No gateway found



```
PC2> ping 192.168.1.10

No gateway found

PC2> ping 192.168.2.40

84 bytes from 192.168.2.40 icmp_seq=1 ttl=64 time=15.281 ms
84 bytes from 192.168.2.40 icmp_seq=2 ttl=64 time=7.287 ms
84 bytes from 192.168.2.40 icmp_seq=3 ttl=64 time=19.246 ms
84 bytes from 192.168.2.40 icmp_seq=4 ttl=64 time=18.490 ms
84 bytes from 192.168.2.40 icmp_seq=5 ttl=64 time=16.633 ms

PC2> ping 192.168.2.60

84 bytes from 192.168.2.60 icmp_seq=1 ttl=64 time=19.731 ms
84 bytes from 192.168.2.60 icmp_seq=2 ttl=64 time=13.454 ms
84 bytes from 192.168.2.60 icmp_seq=3 ttl=64 time=11.300 ms
84 bytes from 192.168.2.60 icmp_seq=4 ttl=64 time=15.381 ms
84 bytes from 192.168.2.60 icmp_seq=5 ttl=64 time=9.182 ms

PC2> ping 192.168.1.30
```

```
PC2> ping 192.168.1.10
```

No gateway found

```
PC2> ping 192.168.2.40
```

```
84 bytes from 192.168.2.40 icmp_seq=1 ttl=64 time=15.281 ms
84 bytes from 192.168.2.40 icmp_seq=2 ttl=64 time=7.287 ms
84 bytes from 192.168.2.40 icmp_seq=3 ttl=64 time=19.246 ms
84 bytes from 192.168.2.40 icmp_seq=4 ttl=64 time=18.490 ms
84 bytes from 192.168.2.40 icmp_seq=5 ttl=64 time=16.633 ms
```

```
PC2> ping 192.168.2.60
```

```
84 bytes from 192.168.2.60 icmp_seq=1 ttl=64 time=19.731 ms
84 bytes from 192.168.2.60 icmp_seq=2 ttl=64 time=13.454 ms
84 bytes from 192.168.2.60 icmp_seq=3 ttl=64 time=11.300 ms
84 bytes from 192.168.2.60 icmp_seq=4 ttl=64 time=15.381 ms
```

```
84 bytes from 192.168.2.60 icmp_seq=5 ttl=64 time=9.182 ms
```

```
PC2> ping 192.168.1.30
```

```
No gateway found
```

```
PC2> ping 192.168.1.50
```

```
No gateway found
```

3) Перехватить в WireShark пакеты с тегами и без тегов (nb!), результаты задокументировать

кадр с тегом vlan 20

Согласно ранее зафиксированному состоянию STP, на SW5

```
Gi0/0 -> Forwarding для VLAN 20  
Gi0/2 -> Forwarding для VLAN 333  
Gi0/1 -> заблокирован  
Gi0/3 -> заблокирован
```

Т.е. для захвата кадров VLAN20 выбрал точкой захвата межкоммутаторный линк от SW5 Gi0/0:

Путь пакета:

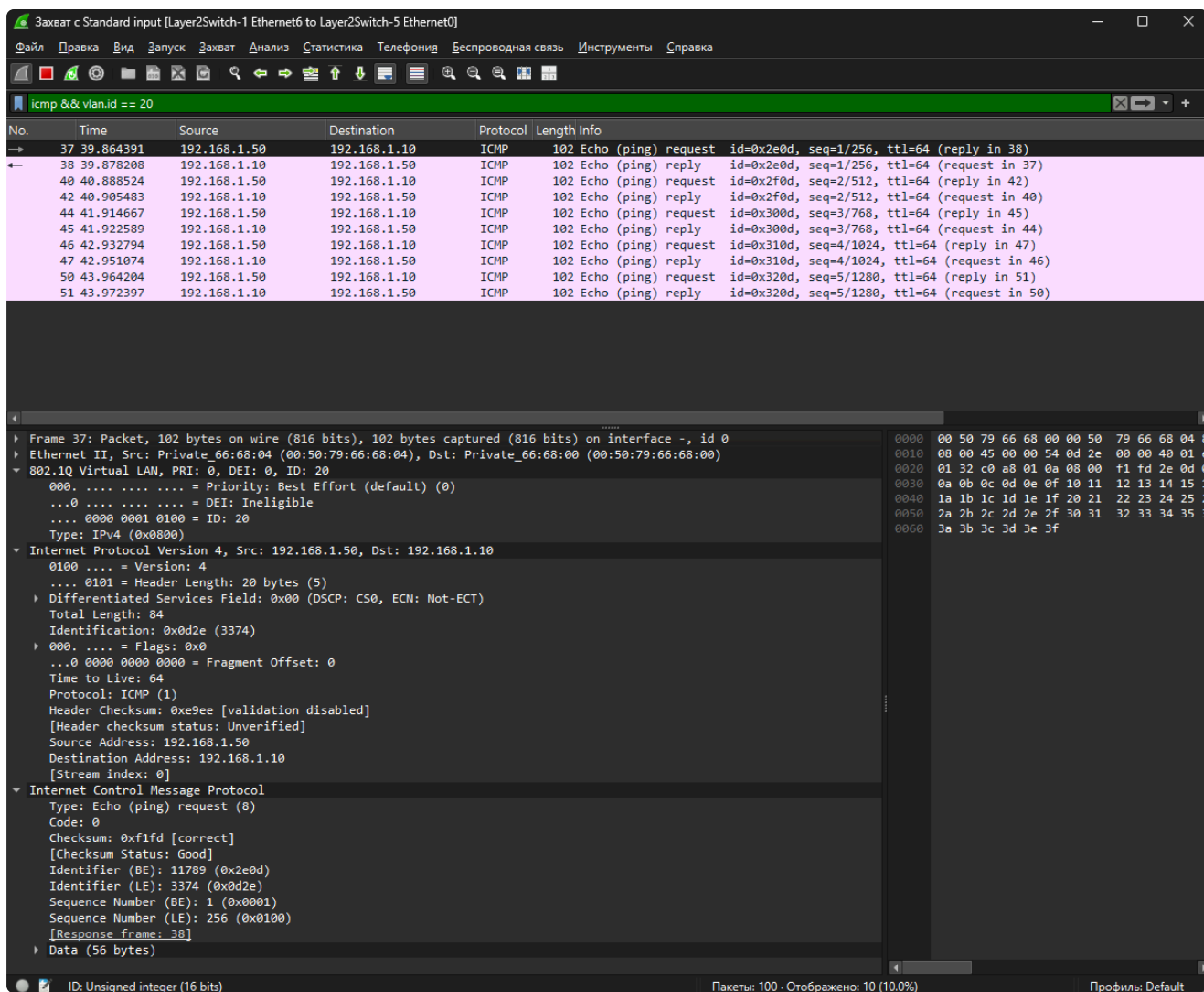
```
PC5 -> SW5 Gi1/0 (access VLAN 20)  
      -> SW5 Gi0/0 (trunk, кадр с тегом VLAN 20)  
      -> SW1 -> другой коммутатор -> PC1 или PC3
```

Пинганул pc1:

```
PC5> ping 192.168.1.30
```

Далее в wireshark:

```
icmp && vlan.id == 20
```



-> видим раздел 802.1Q virtual LAN, PRI: 0, DEI: 0, ID: 20

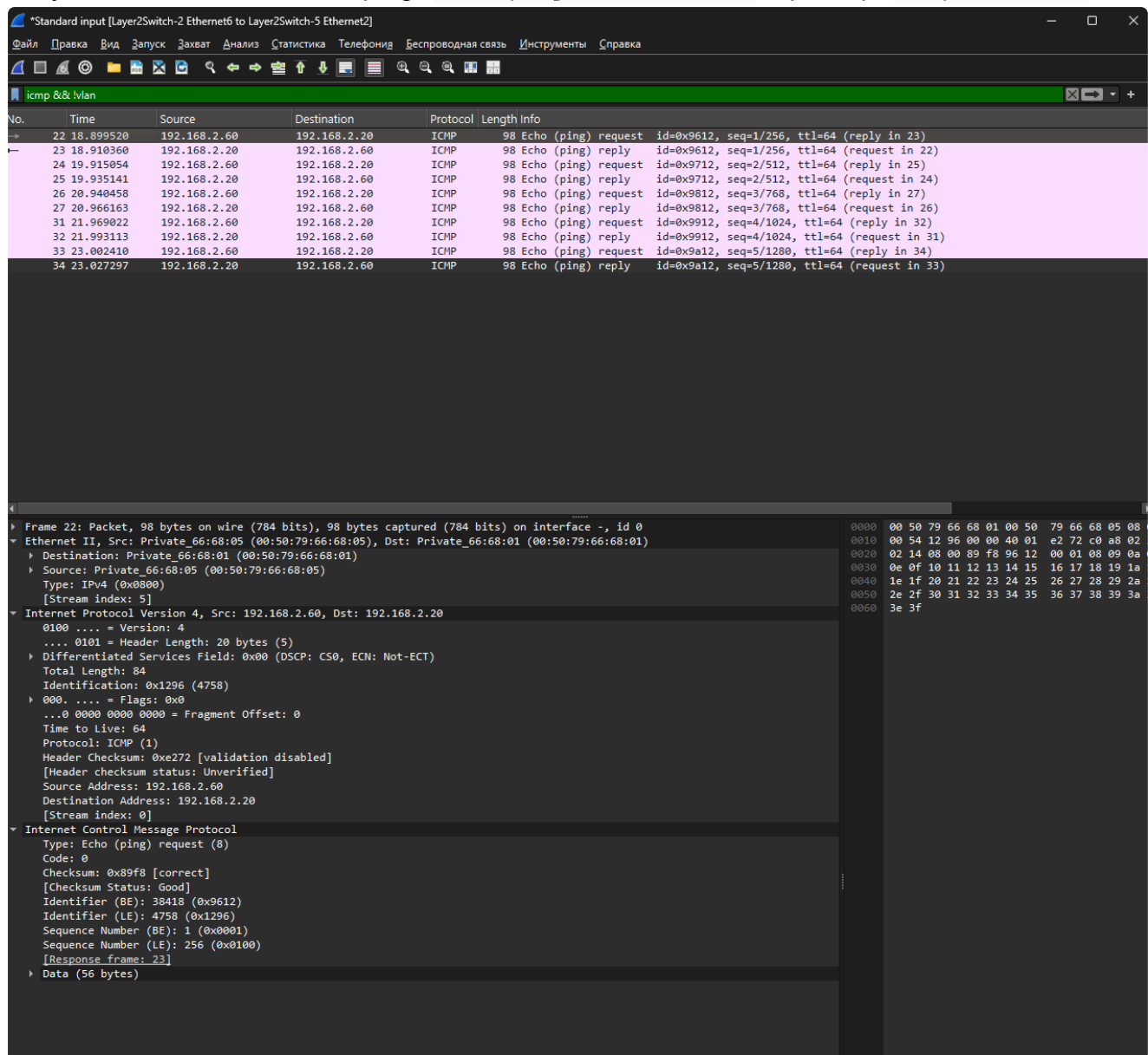
Это доказывает, что VLAN 20 передаётся через trunk с тегом 802.1Q.

кадр без тега native vlan 333

точка захвата -> межкоммутаторный линк от SW5 Gi0/2, путь пакета:

```
PC6 -> SW5 Gi1/1 (access VLAN 333)
-> SW5 Gi0/2 (trunk, native VLAN 333 без тега)
-> SW2 -> другой коммутатор -> PC2 или PC4
```

запустил захват, выполнил ping: PC6> ping 192.168.2.20 , фильтр: icmp && !vlan



Wireshark packet capture showing ICMP ping requests and replies between 192.168.2.20 and 192.168.2.60. The filter is 'icmp && !vlan'. The packet list shows 13 packets (No. 22-34). The packet details pane shows the structure of packet 22, including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (Echo (ping) request).

No.	Time	Source	Destination	Protocol	Length	Info
22	18.899520	192.168.2.60	192.168.2.20	ICMP	98	Echo (ping) request id=0x9612, seq=1/256, ttl=64 (reply in 23)
23	18.910360	192.168.2.20	192.168.2.60	ICMP	98	Echo (ping) reply id=0x9612, seq=1/256, ttl=64 (request in 22)
24	19.915054	192.168.2.60	192.168.2.20	ICMP	98	Echo (ping) request id=0x9712, seq=2/512, ttl=64 (reply in 25)
25	19.935141	192.168.2.20	192.168.2.60	ICMP	98	Echo (ping) reply id=0x9712, seq=2/512, ttl=64 (request in 24)
26	20.940458	192.168.2.60	192.168.2.20	ICMP	98	Echo (ping) request id=0x9812, seq=3/768, ttl=64 (reply in 27)
27	20.966163	192.168.2.20	192.168.2.60	ICMP	98	Echo (ping) reply id=0x9812, seq=3/768, ttl=64 (request in 26)
31	21.969022	192.168.2.60	192.168.2.20	ICMP	98	Echo (ping) request id=0x9912, seq=4/1024, ttl=64 (reply in 32)
32	21.993113	192.168.2.20	192.168.2.60	ICMP	98	Echo (ping) reply id=0x9912, seq=4/1024, ttl=64 (request in 31)
33	23.002410	192.168.2.60	192.168.2.20	ICMP	98	Echo (ping) request id=0x9a12, seq=5/1280, ttl=64 (reply in 34)
34	23.027297	192.168.2.20	192.168.2.60	ICMP	98	Echo (ping) reply id=0x9a12, seq=5/1280, ttl=64 (request in 33)

Frame 22: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:05 (00:50:79:66:68:05), Dst: Private_66:68:01 (00:50:79:66:68:01)
Destination: Private_66:68:01 (00:50:79:66:68:01)
Source: Private_66:68:05 (00:50:79:66:68:05)
Type: IPv4 (0x0800)
[Stream index: 5]
Internet Protocol Version 4, Src: 192.168.2.60, Dst: 192.168.2.20
0100 = Version: 4
... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0x1296 (4758)
000. = Flags: 0x0
...0 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0xe272 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.2.60
Destination Address: 192.168.2.20
[Stream index: 0]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x89f8 [correct]
[Checksum Status: Good]
Identifier (BE): 38418 (0x9612)
Identifier (LE): 4758 (0x1296)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[Response frame: 23]
Data (56 bytes)

-> видим что в icmp кадрах нету раздела 802.1Q.

-> число 333 внутри кадра тоже не видим, так как принимающий коммутатор относит нетегированный кадр к vlan 333 на основании настройки native vlan 333.

4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

-> Сохранено

5*) Опциональное задание: Добавить в схему маршрутизатор, подключенный к коммутаторам Layer2Switch1 и Layer2Switch2, настроить через него маршрутизацию между VLAN

Полезная информация: избыточные физические каналы можно поместить в отдельные VLAN и обойтись без STP